

Diagnosing Sacral Somatic Dysfunction – Work Slides

Michael J. Terzella, D.O.

Department of Osteopathic Manipulative Medicine

mterzell@nyit.edu



Do.
Make.
Heal.

Innovate.

Reinvent the Future.

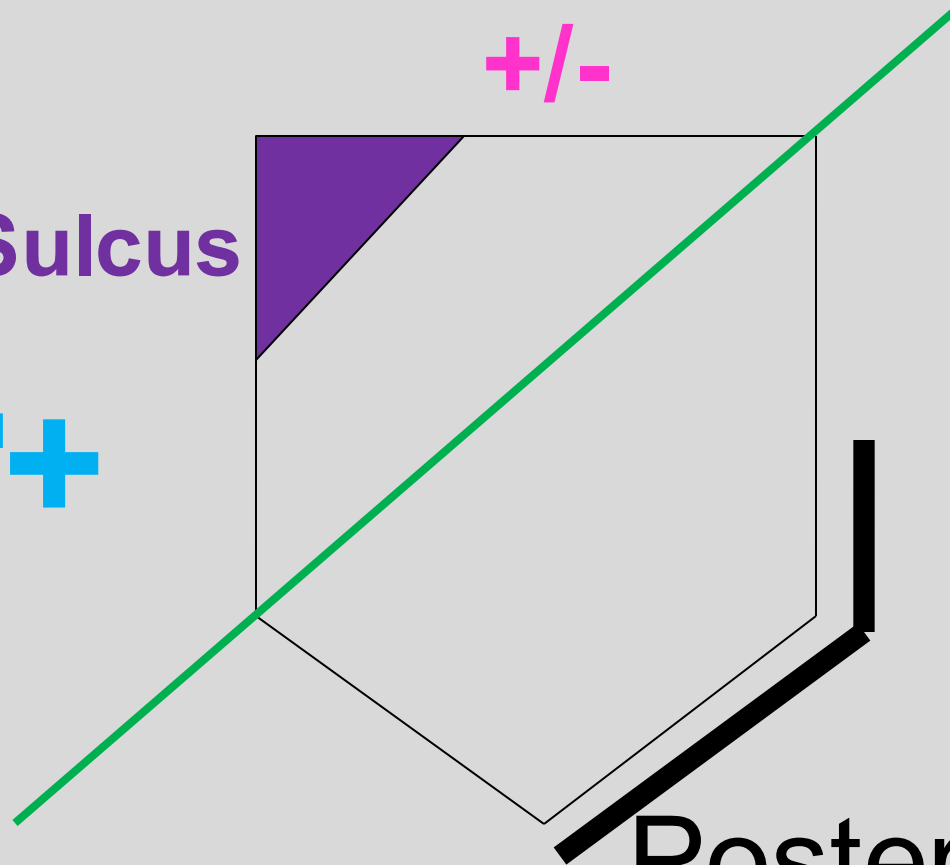
Diagramming Sacral Diagnosis

Side of Lumbar Rotation +
Spring test/backward bending (sphinx) test
+/-

Axis

Deep Sulcus

Positive Seated
Flexion test +



Posterior-inferior
ILA

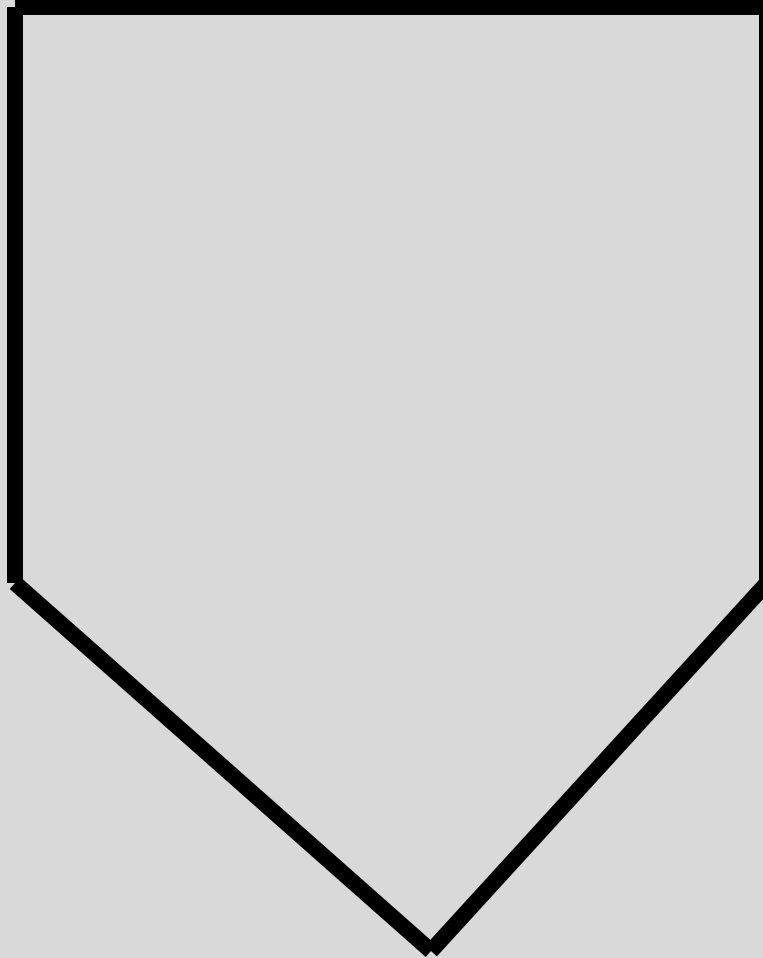
- Using the notation from the previous slide, diagram the following sacra according to the physical exam findings in the left lower corners.
- Then try to figure out the diagnosis
- All this will be demonstrated in lecture so if you are having trouble, don't worry.

- Also note, the following slides do not mention an L5 diagnosis.
- If L5 is rotated the same direction as the sacrum, then the diagnosis is a “sacral rotation”
 - Example: right on right forward sacral **rotation**
- If L5 is rotated the opposite direction as the sacrum **OR** if there is no L5 dysfunction, then the diagnosis is a “sacral torsion”
 - Example: right on right forward sacral **torsion**

- Just to be clear:
Work Slides 1 – 10 do not mention an L5 diagnosis. You will need L5 to determine a torsion vs. a rotation. The following examples are here for you to work out the sacral part of the diagnosis.
When studying, add in the L5 diagnosis to make it more interesting.
- **Work slides 11 & 12 do mention an L5 diagnosis so you can practice making a sacral diagnosis given ONLY the L5 diagnosis.**
- Also, the abbreviation “bbt”, which is NOT an officially accepted abbreviation, is used for “backward bending test” on the following slides.

Work slide #1

L5
diagnosis?

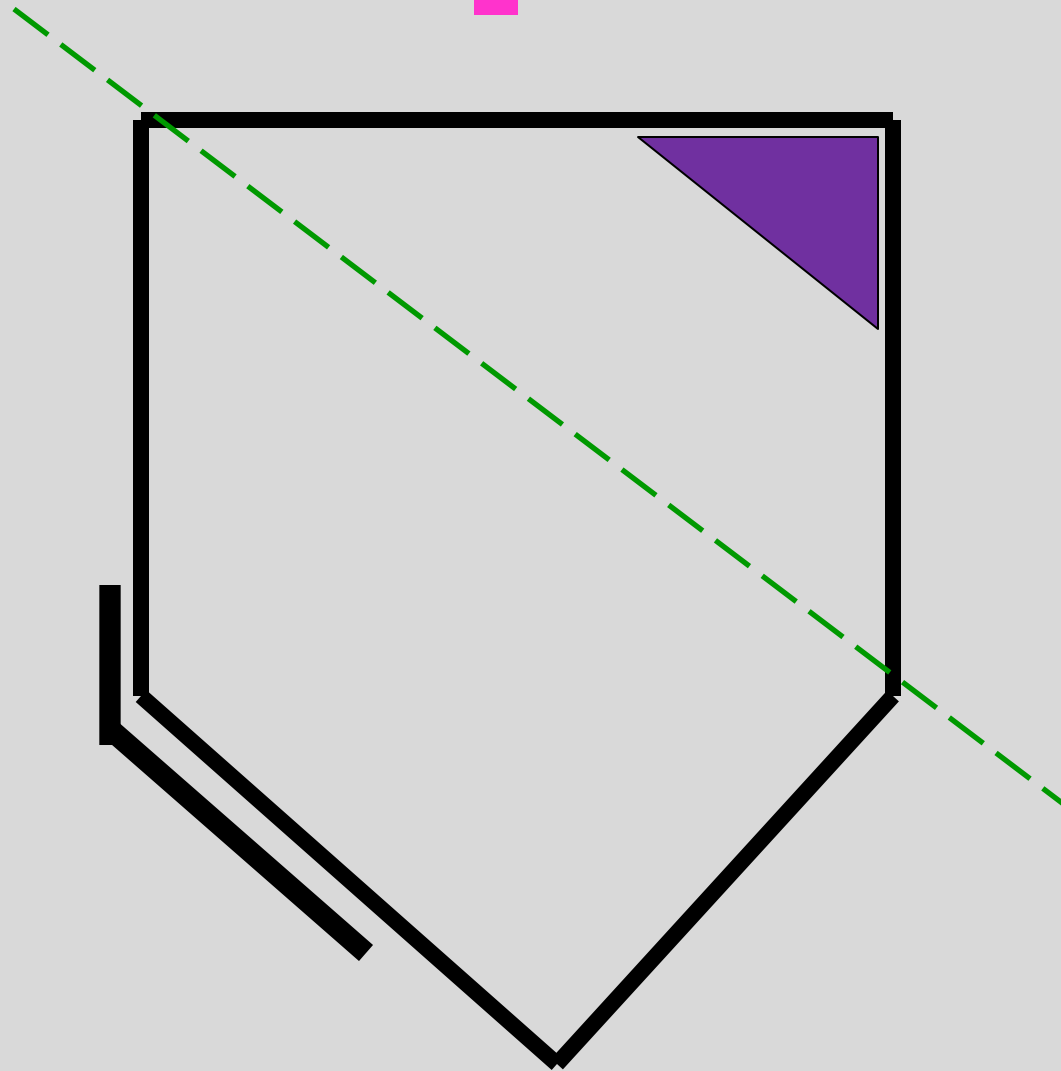


- Negative Spring test
- Neg. bbt /sphinx test
- Deep sulcus on right
- Posterior/Inferior ILA on the left
- Seated Flexion???

Work slide #1:

Left on Left Forward sacrum

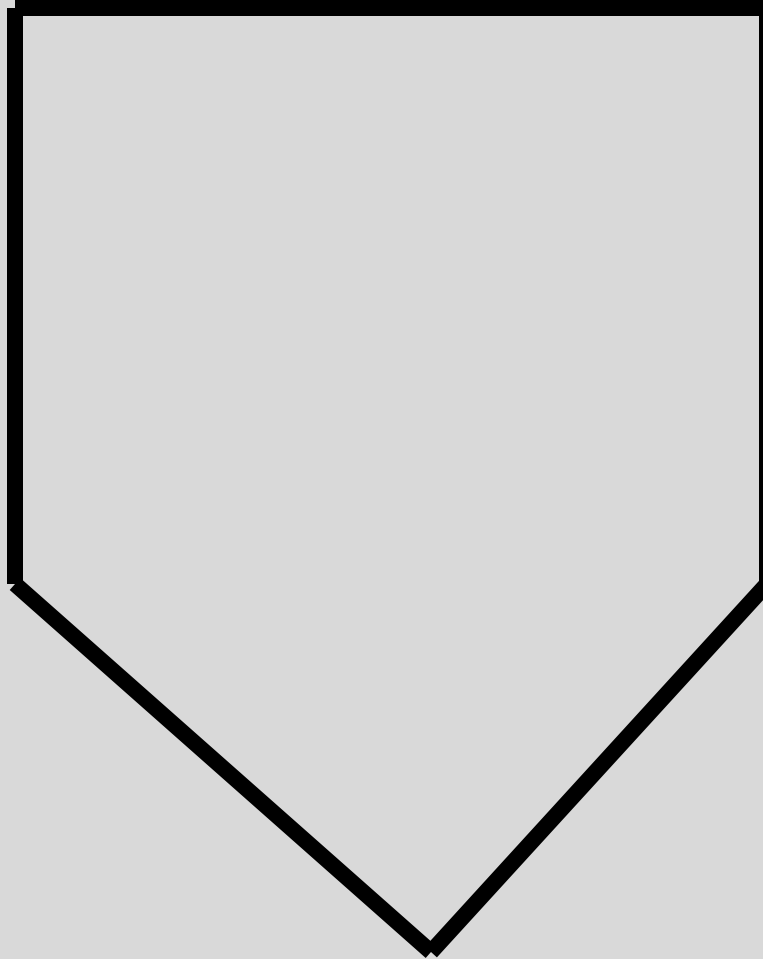
L5
diagnosis?



- Negative Spring test
- Neg. bbt /sphinx test
- Deep sulcus on right
- Posterior/Inferior ILA on the left
- Seated Flexion???

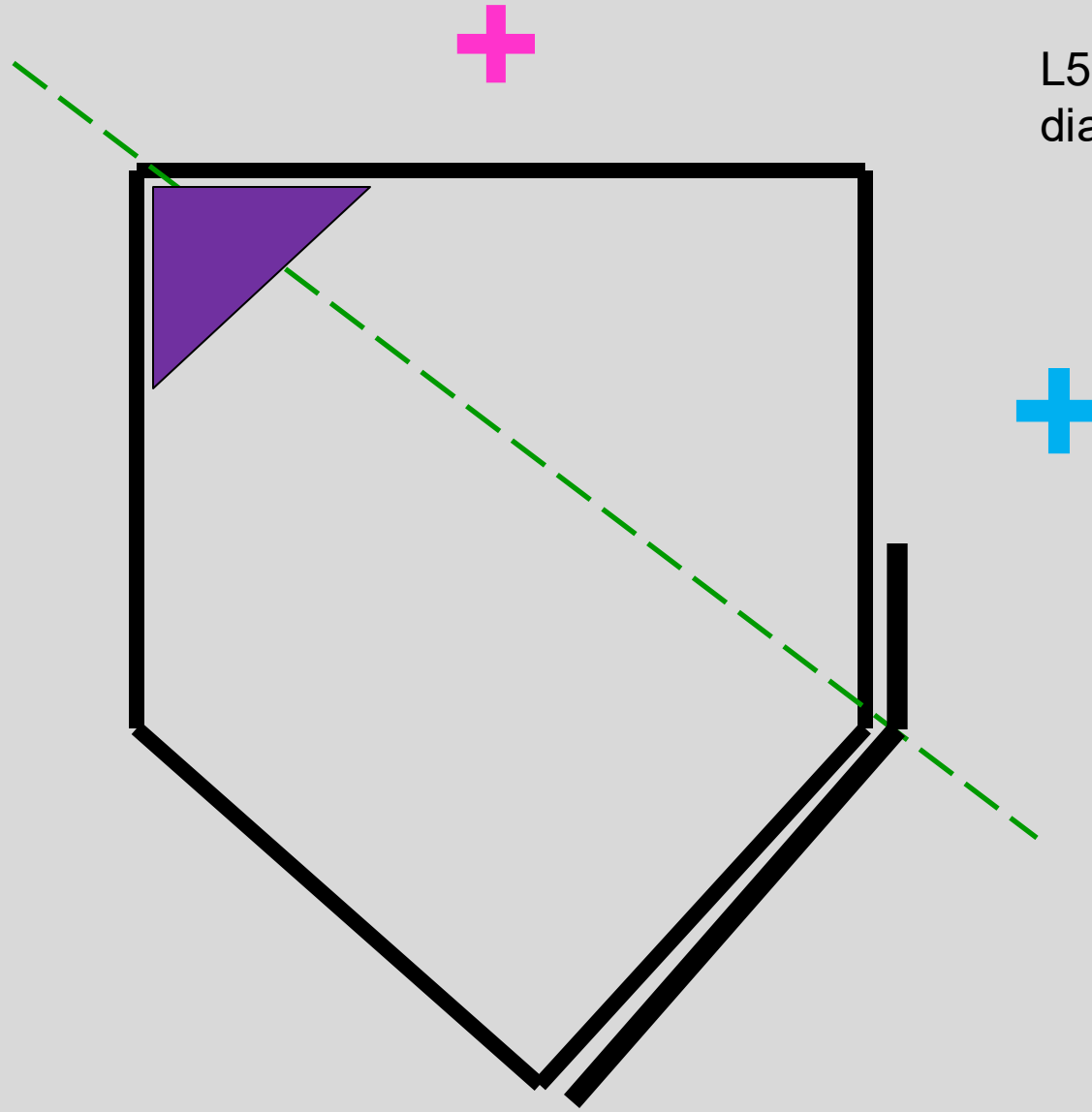
Work slide #2

L5
diagnosis?



- Positive Spring test
- Pos. bbt / sphinx test
- Deep sulcus on left
- Posterior/Inferior ILA on the right
- Seated flexion test??

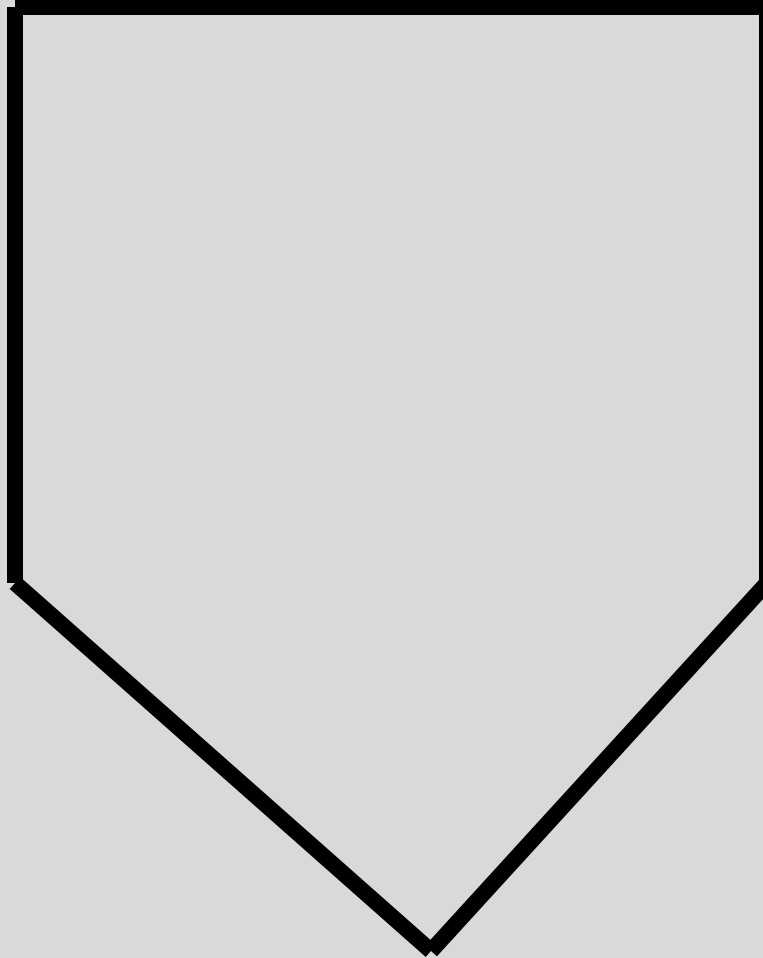
L5
diagnosis?



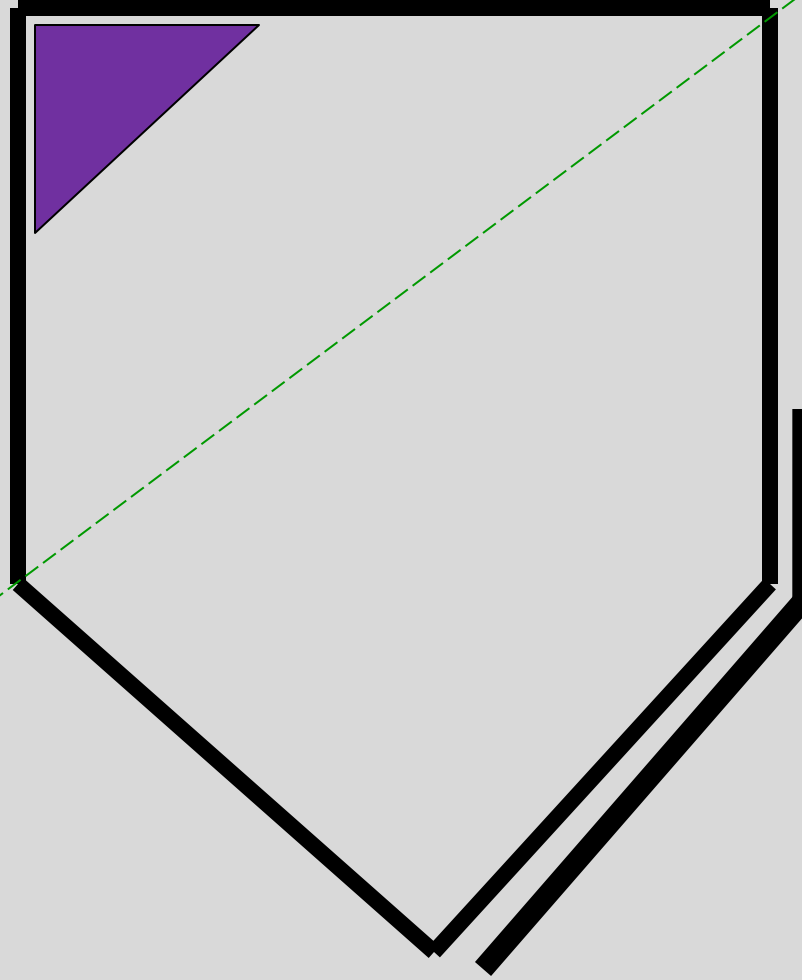
- Positive Spring test
- Pos. bbt / sphinx test
- Deep sulcus on left
- Posterior/Inferior ILA on the right
- Seated flexion test??

Work slide #3

L5
diagnosis?



- Negative Spring test
- Neg. bbt / sphinx test
- Seated flexion test??
- Deep sulcus on left
- Posterior/Inferior ILA on the right



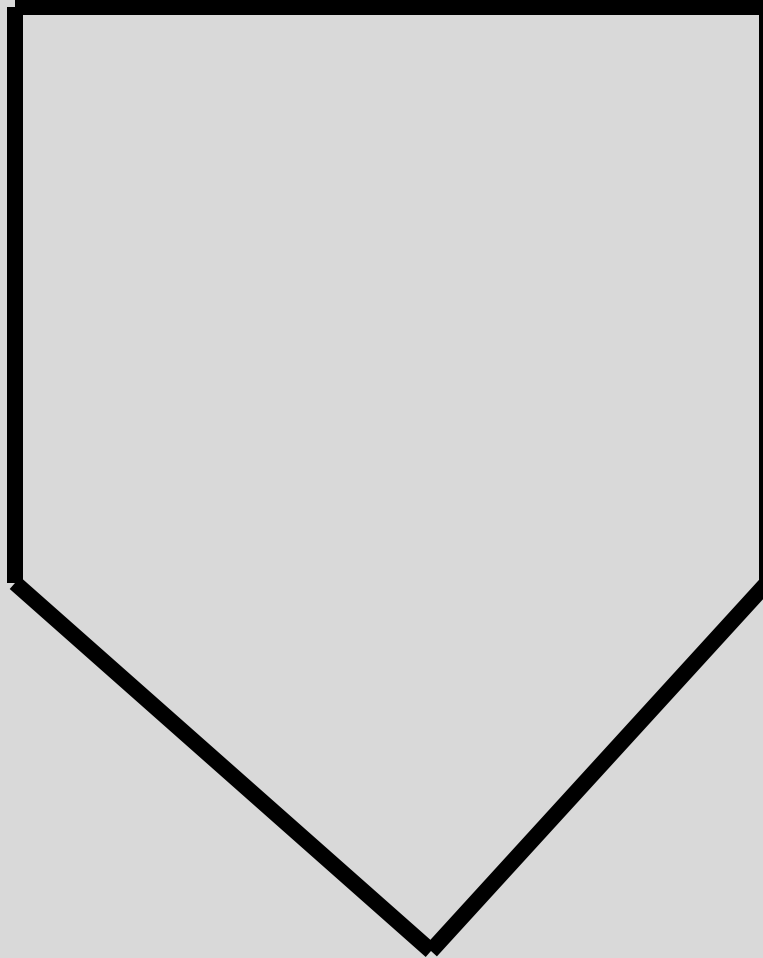
L5
diagnosis?

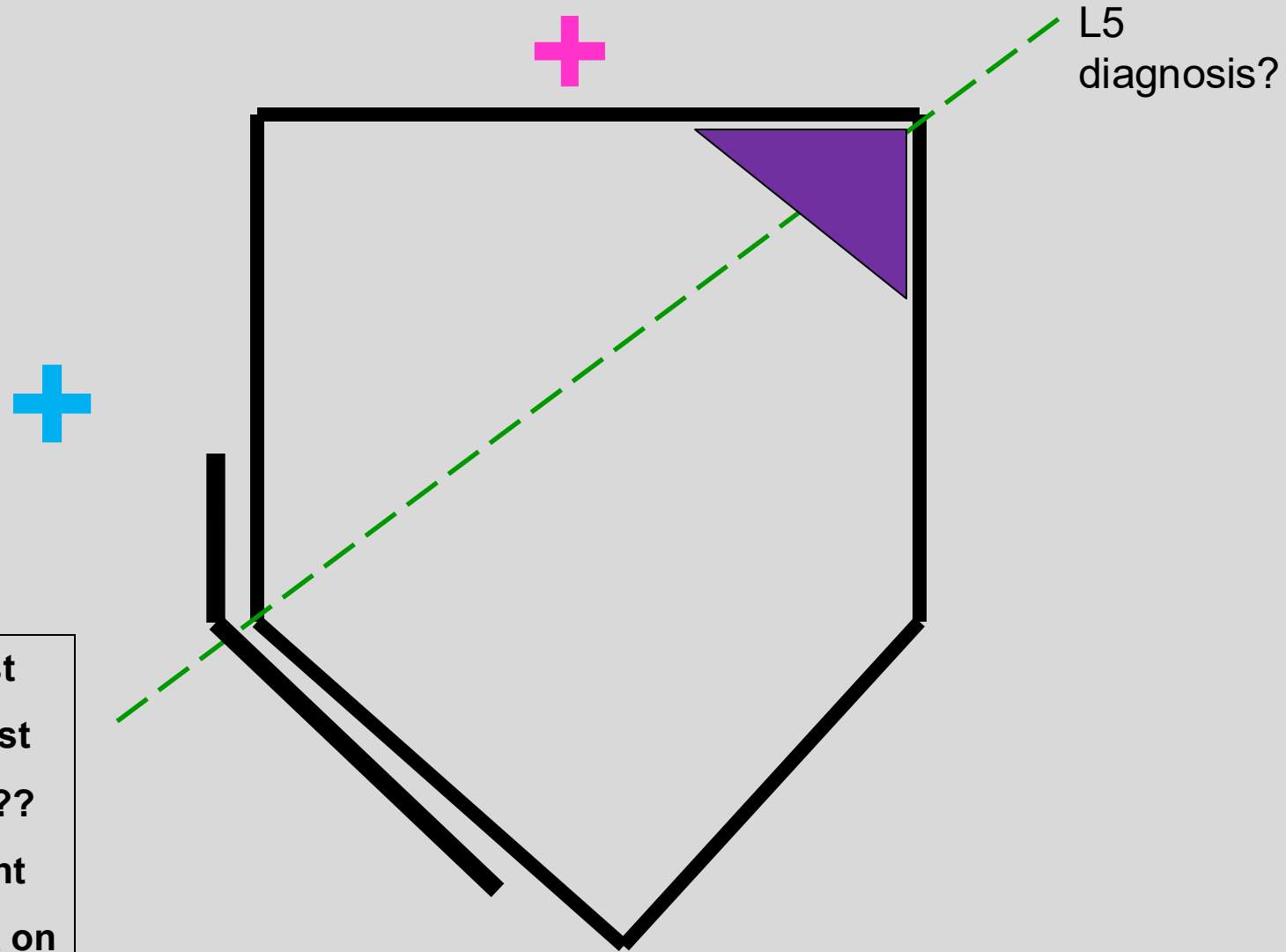
- Negative Spring test
- Neg. bbt / sphinx test
- Seated flexion test??
- Deep sulcus on left
- Posterior/Inferior ILA on the right

Work slide #4

L5
diagnosis?

- Positive Spring test
- Pos. bbt / sphinx test
- Seated Flexion test??
- Deep sulcus on right
- Posterior/Inferior ILA
on the left



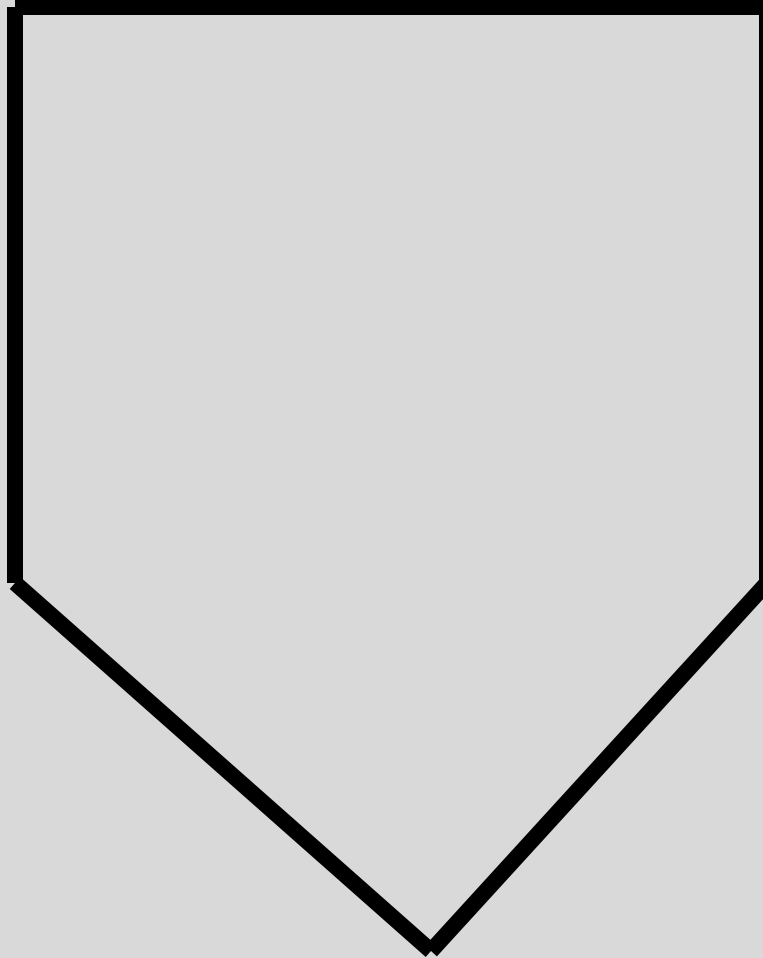


- Positive Spring test
- Pos. bbt / sphinx test
- Seated Flexion test??
- Deep sulcus on right
- Posterior/Inferior ILA on the left

Work slide #5

L5
diagnosis?

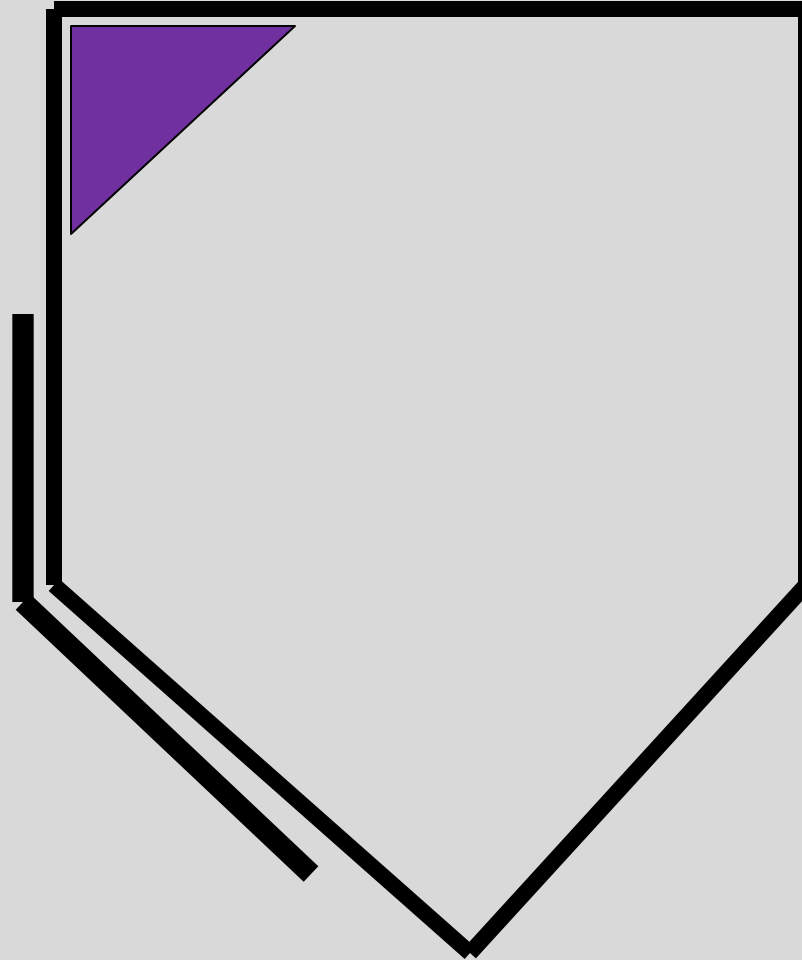
- positive seated flexion test on left
- Neg. spring test
- Neg. bbt /sphinx test
- Deep sulcus on left
- Post. /inferior ILA on left



L5
diagnosis?



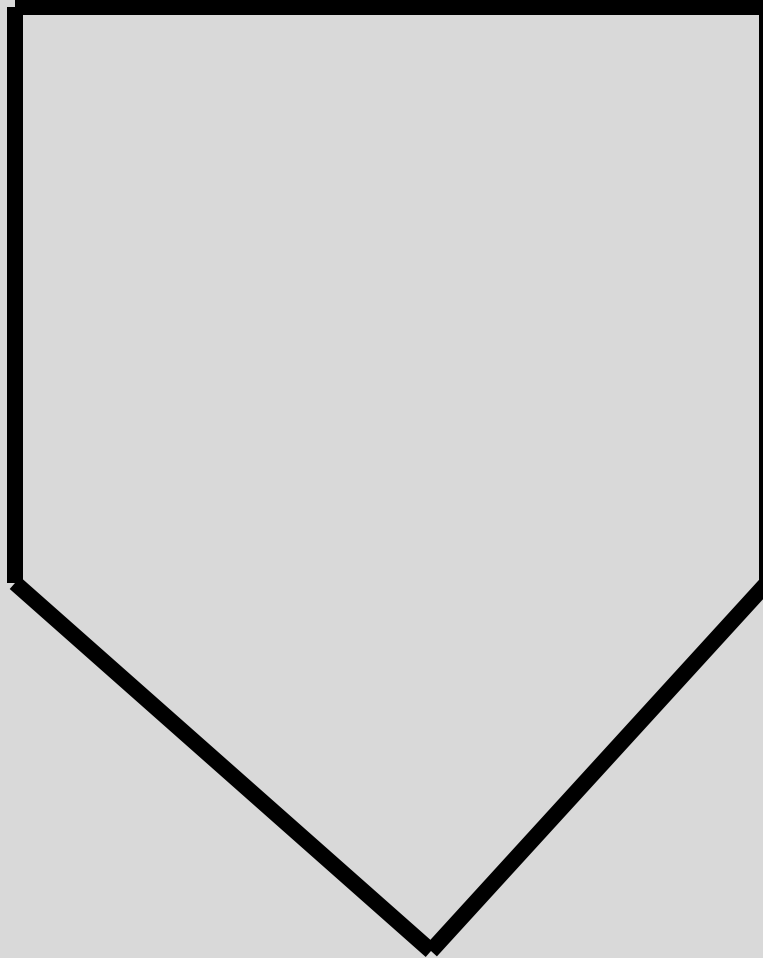
- positive seated flexion test on left
- Neg. spring test
- Neg. bbt /sphinx test
- Deep sulcus on left
- Post. /inferior ILA on left



Work slide #6

L5
diagnosis?

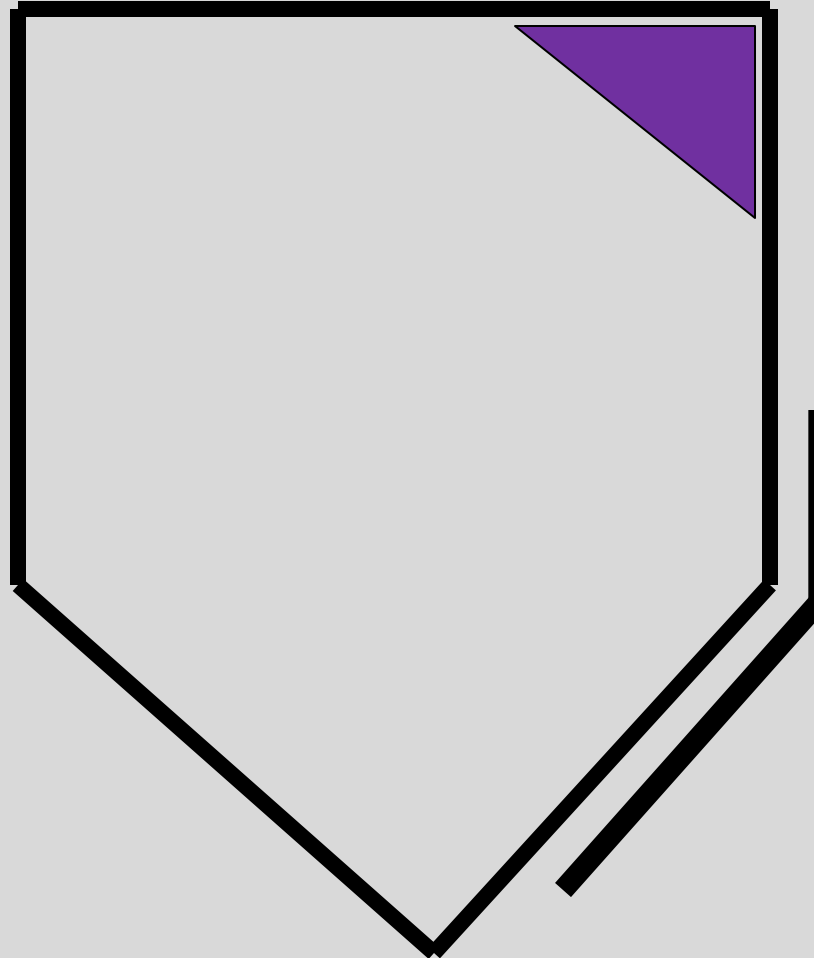
- positive seated flexion test on right
- Neg. spring test
- Neg. bbt / sphinx test
- Deep sulcus on right
- Post. /inferior ILA on right



L5
diagnosis?



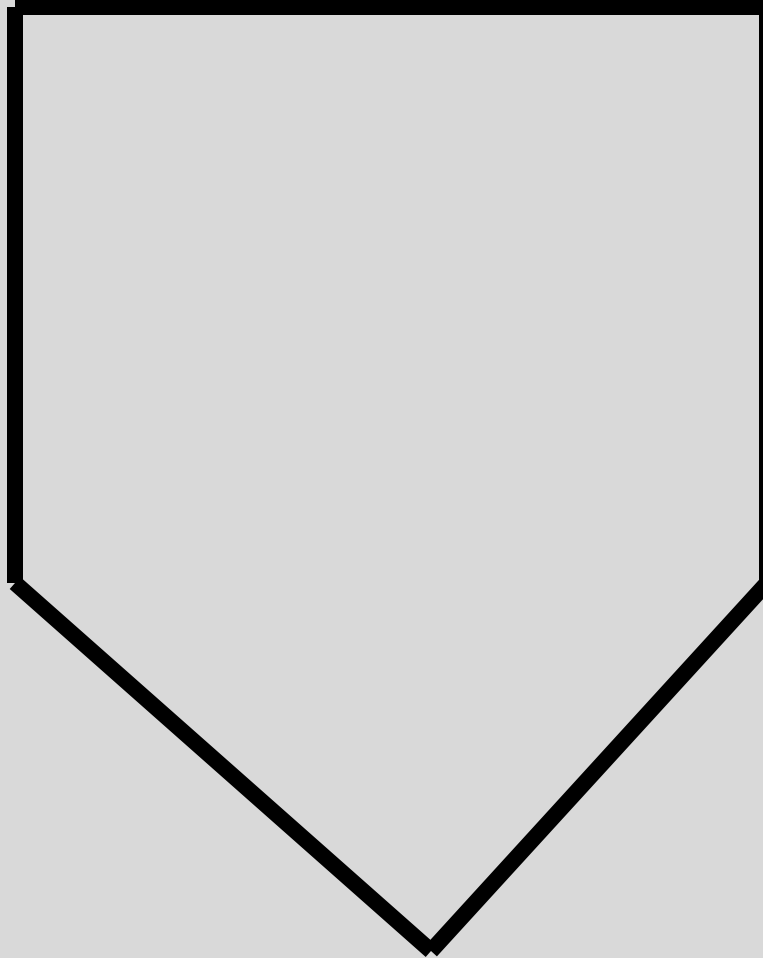
- positive seated flexion test on right
- Neg. spring test
- Neg. bbt / sphinx test
- Deep sulcus on right
- Post. /inferior ILA on right



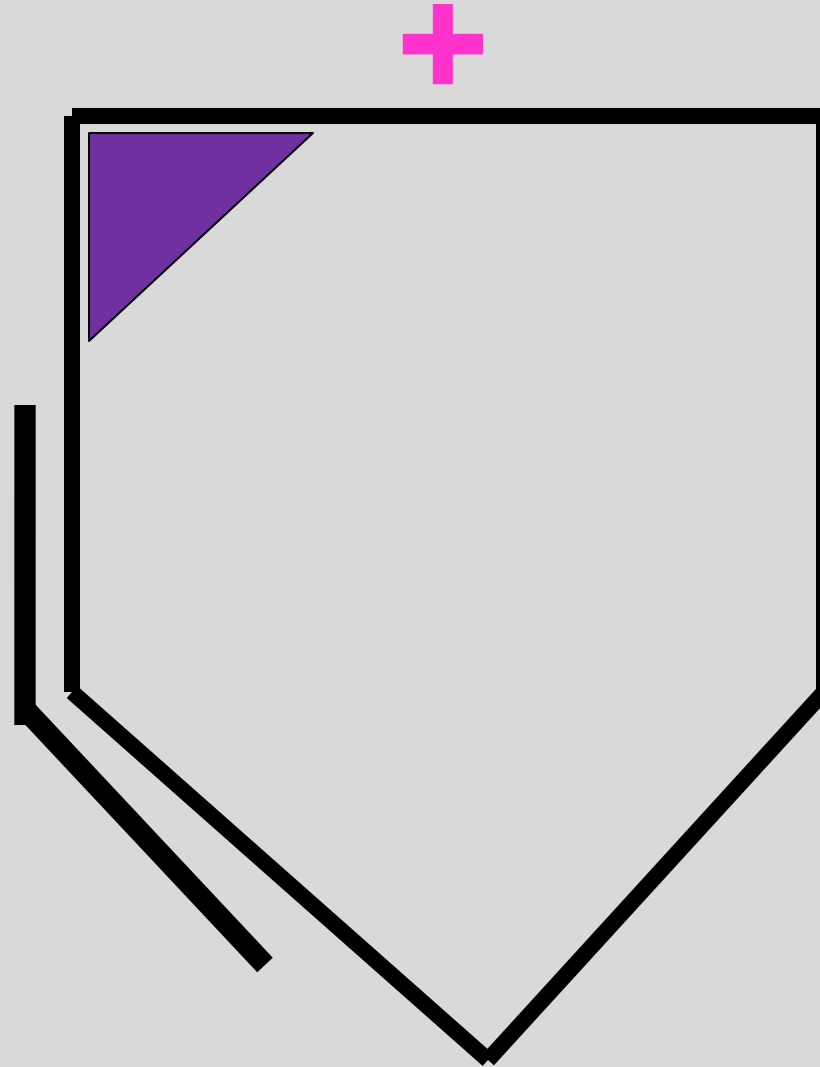
Work slide #7

L5
diagnosis?

- shallow sulcus on the right
- anterior/superior ILA on right
- positive seated flexion test on right
- Pos. bbt / sphinx test
- Pos. spring test



L5
diagnosis?



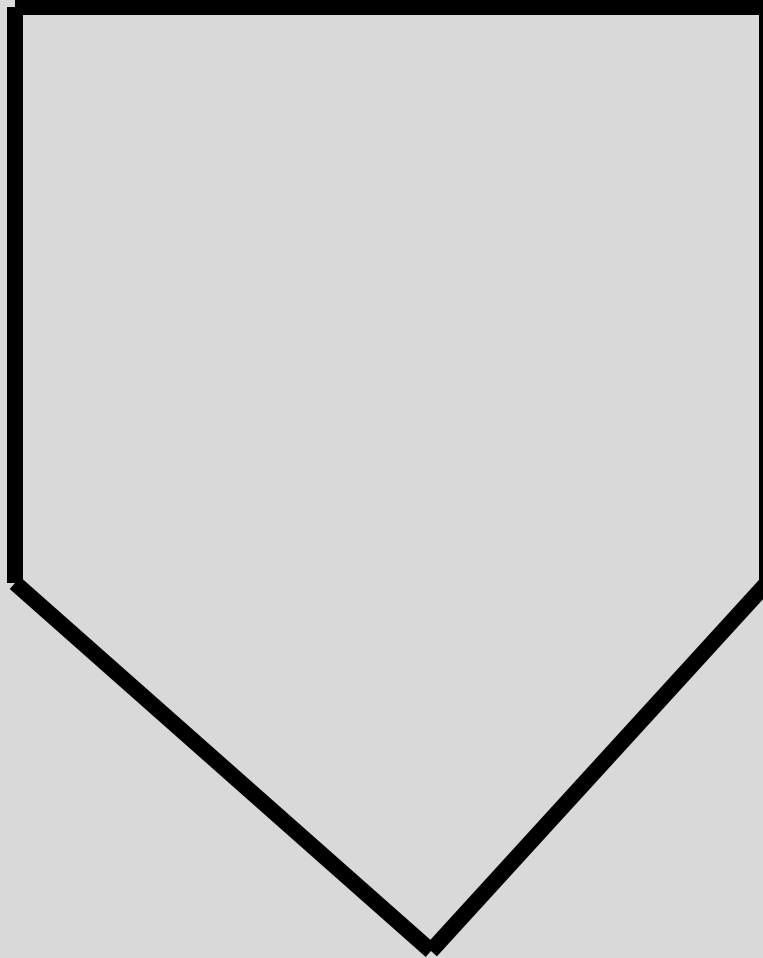
- shallow sulcus on the right
- anterior/superior ILA on right
- positive seated flexion test on right
- Pos. bbt / sphinx test
- Pos. spring test

MAY ALSO BE GIVEN AS:

- Deep sulcus on the left
- Posterior/Inferior ILA on left
- positive seated flexion test on right
- Pos. sphinx test
- Pos. spring test

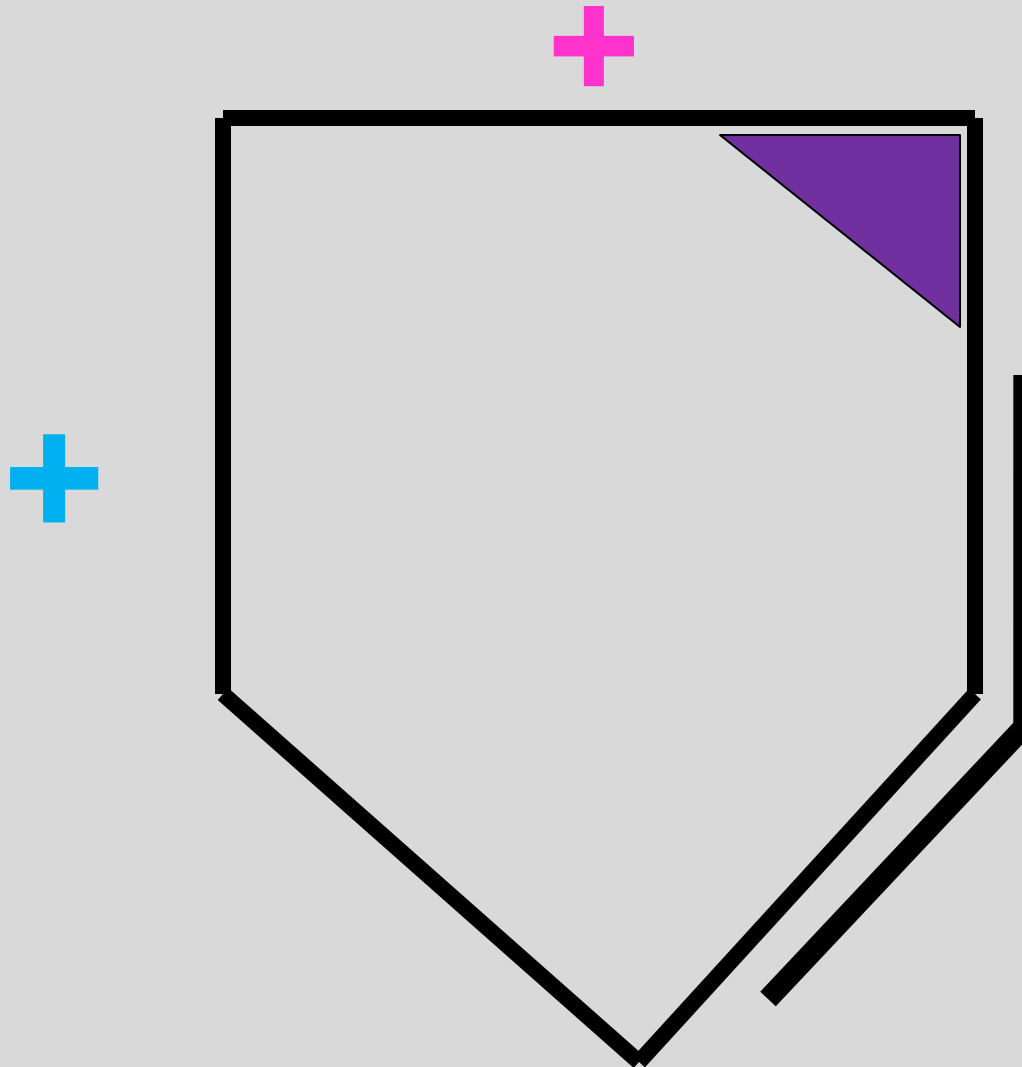
Work slide #8

L5
diagnosis?



- shallow sulcus on the left
- anterior/superior ILA on left
- positive seated flexion test on left
- Pos. bbt / sphinx test
- Pos. spring test

L5
diagnosis?



- shallow sulcus on the left
- anterior/superior ILA on left
- positive seated flexion test on left
- Pos. bbt / sphinx test
- Pos. spring test

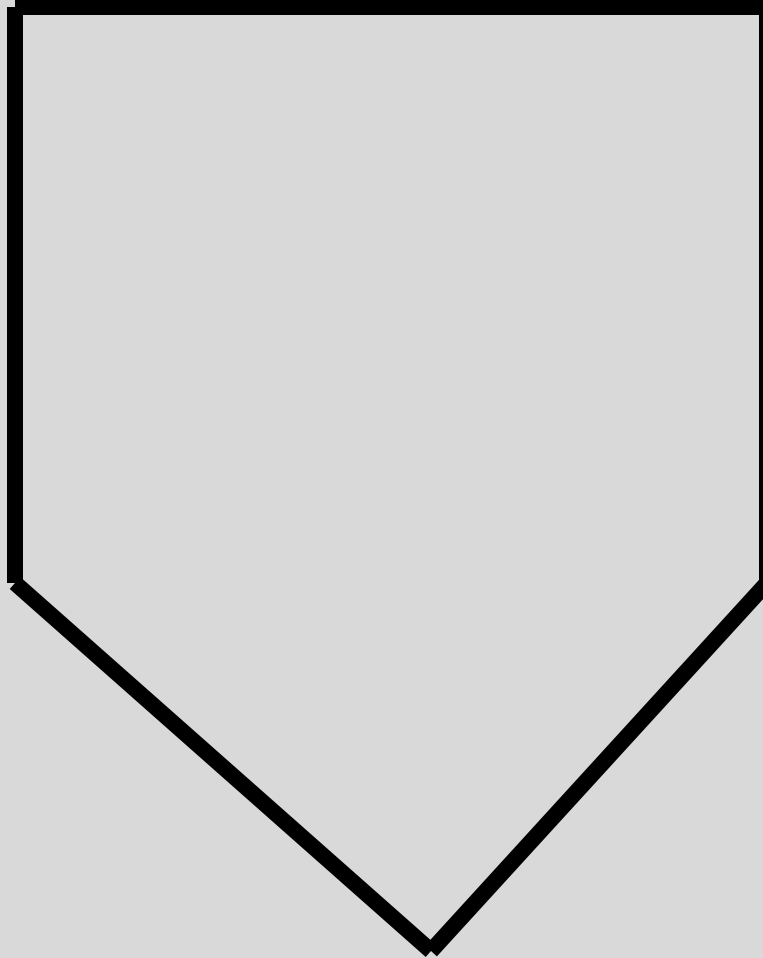
MAY ALSO BE GIVEN AS:

- Deep sulcus on the right
- Posterior/Inferior ILA on right
- positive seated flexion test on left
- Pos. sphinx test
- Pos. spring test

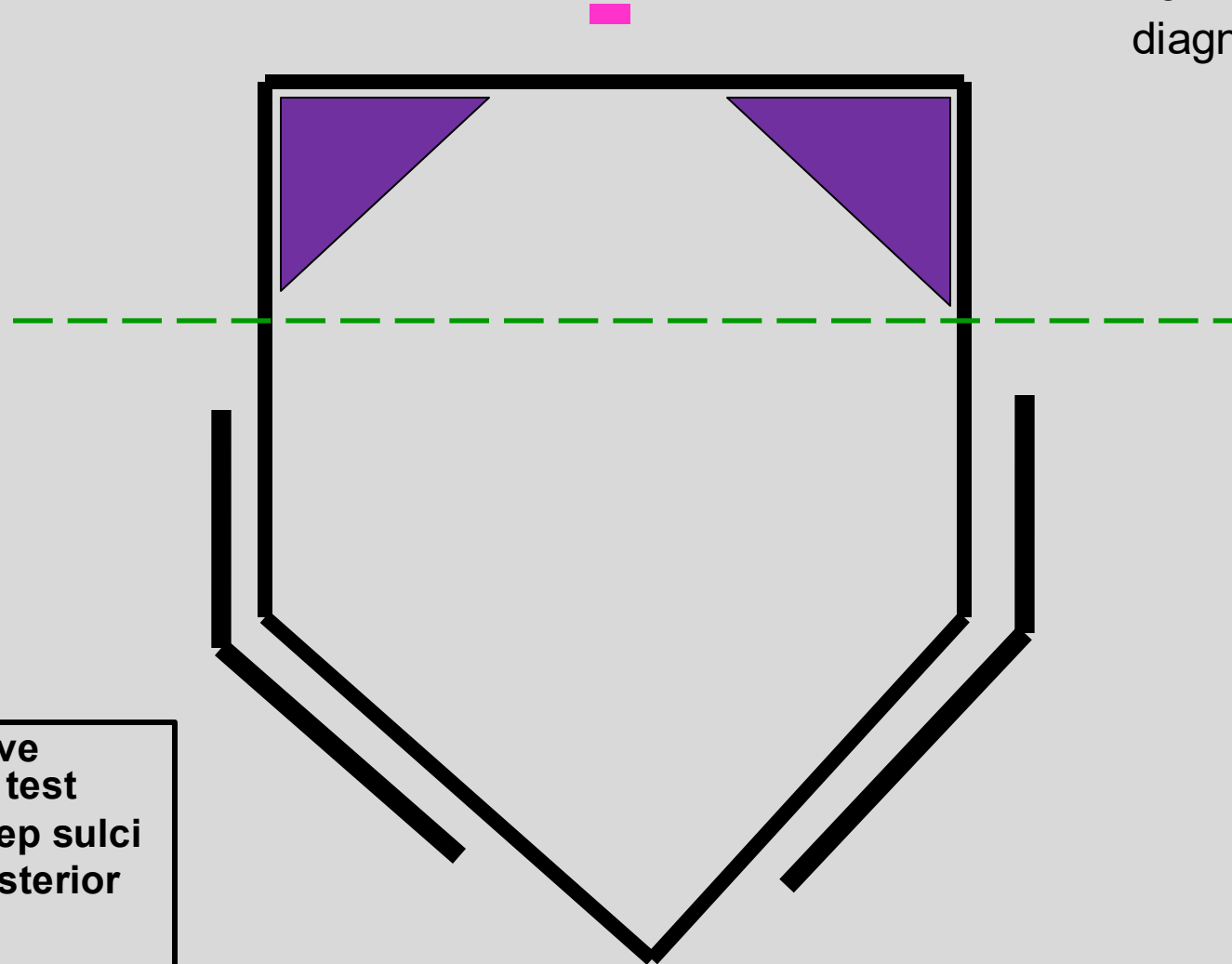
Work slide #9

L5
diagnosis?

- Negative spring test
- B/L deep sulci
- B/L posterior ILAs



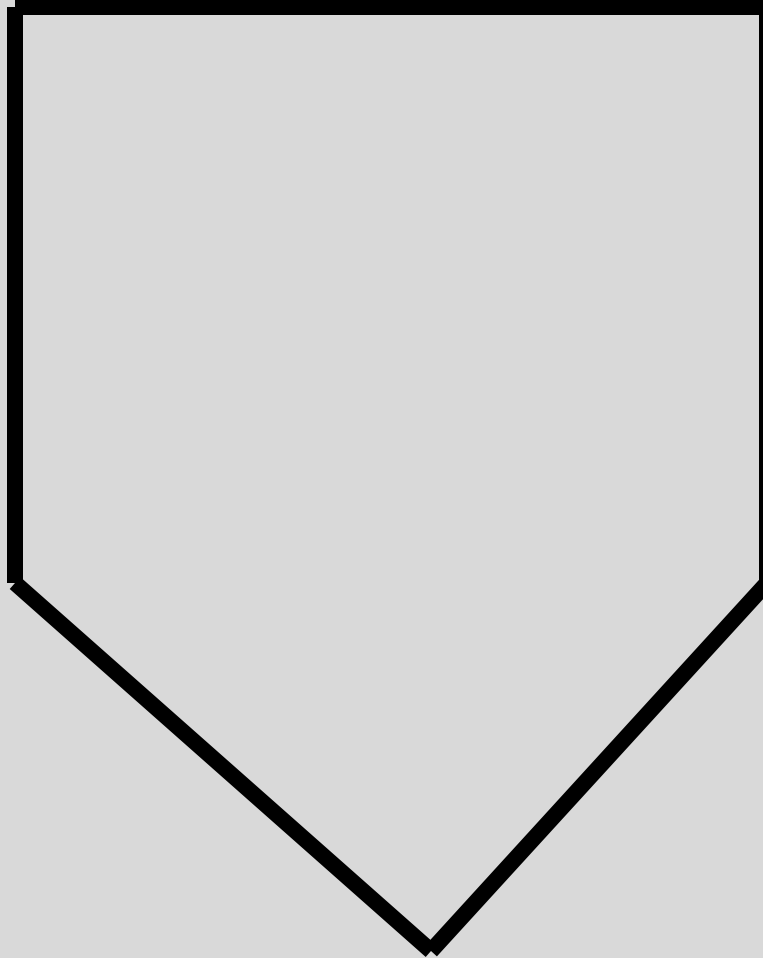
L5
diagnosis?



- Negative spring test
- B/L deep sulci
- B/L posterior ILAs

Work slide #10

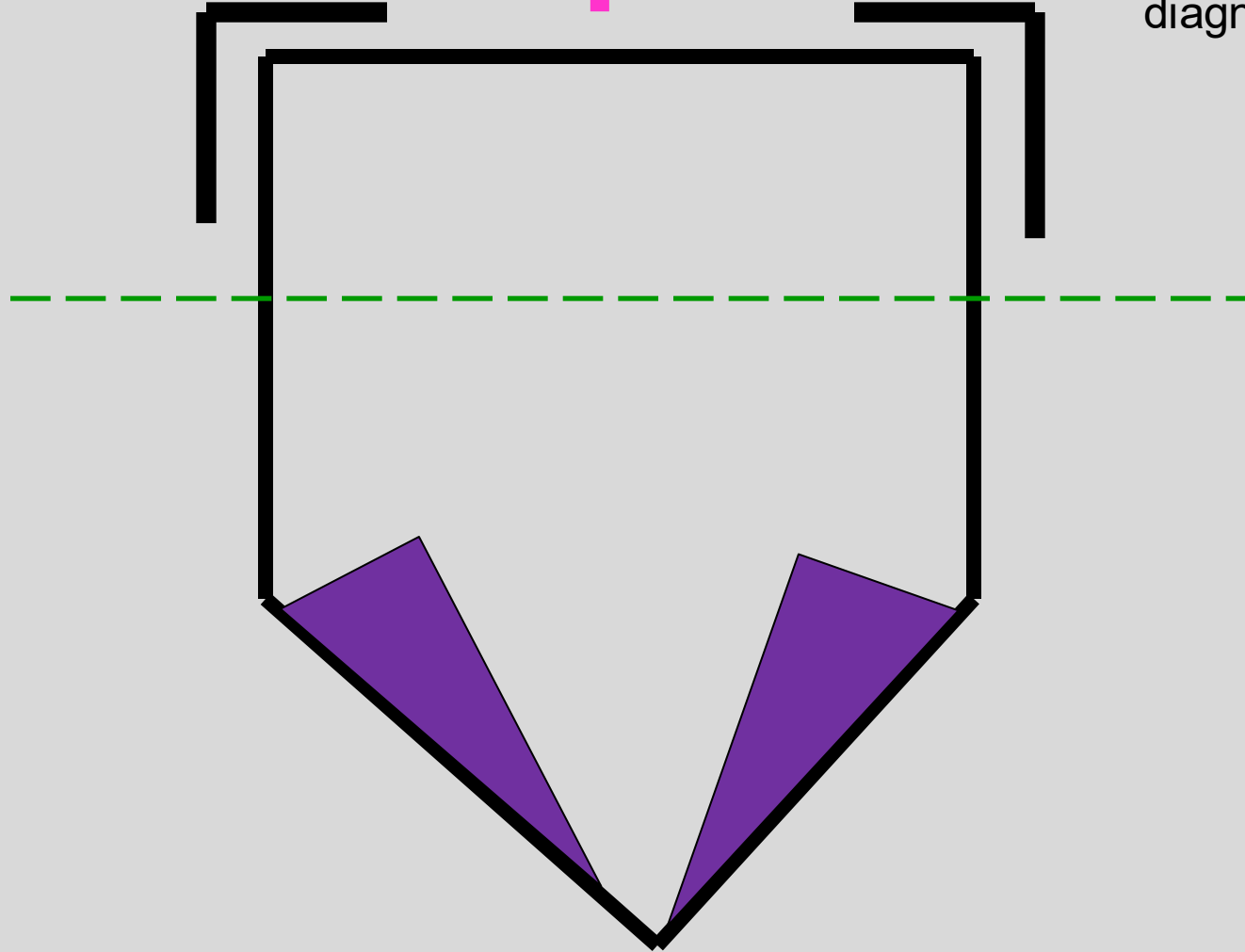
L5
diagnosis?



- positive spring test
- B/L shallow sulci
- B/L anterior ILAs



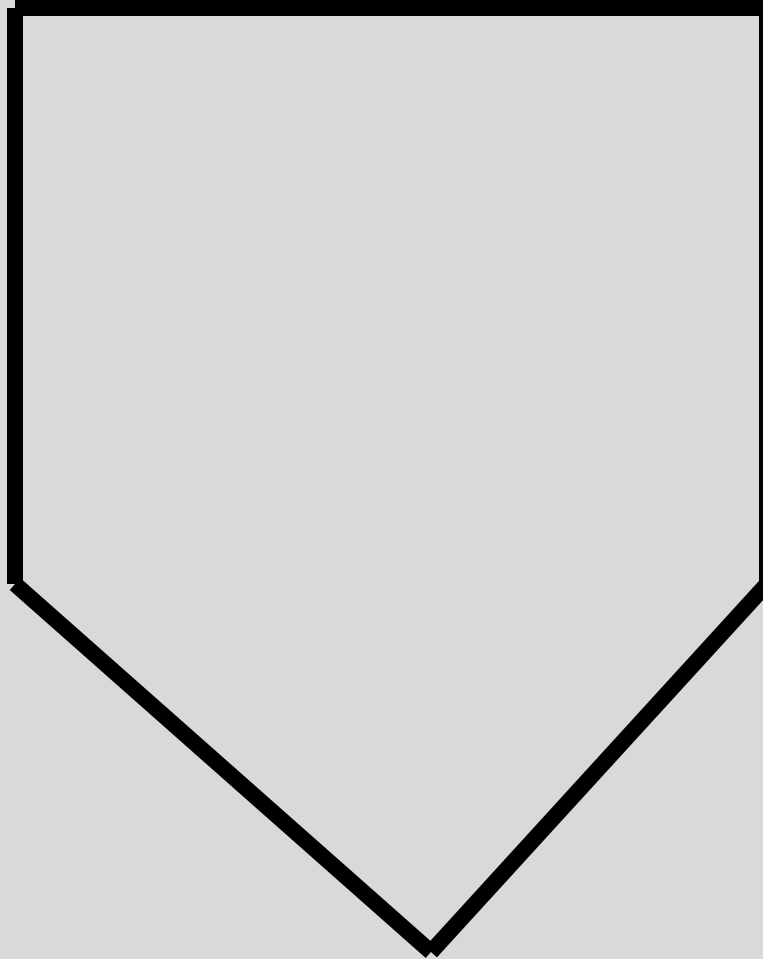
L5
diagnosis?



- positive spring test
- B/L shallow sulci
- B/L anterior ILAs

Work slide #11

L5 FRS left

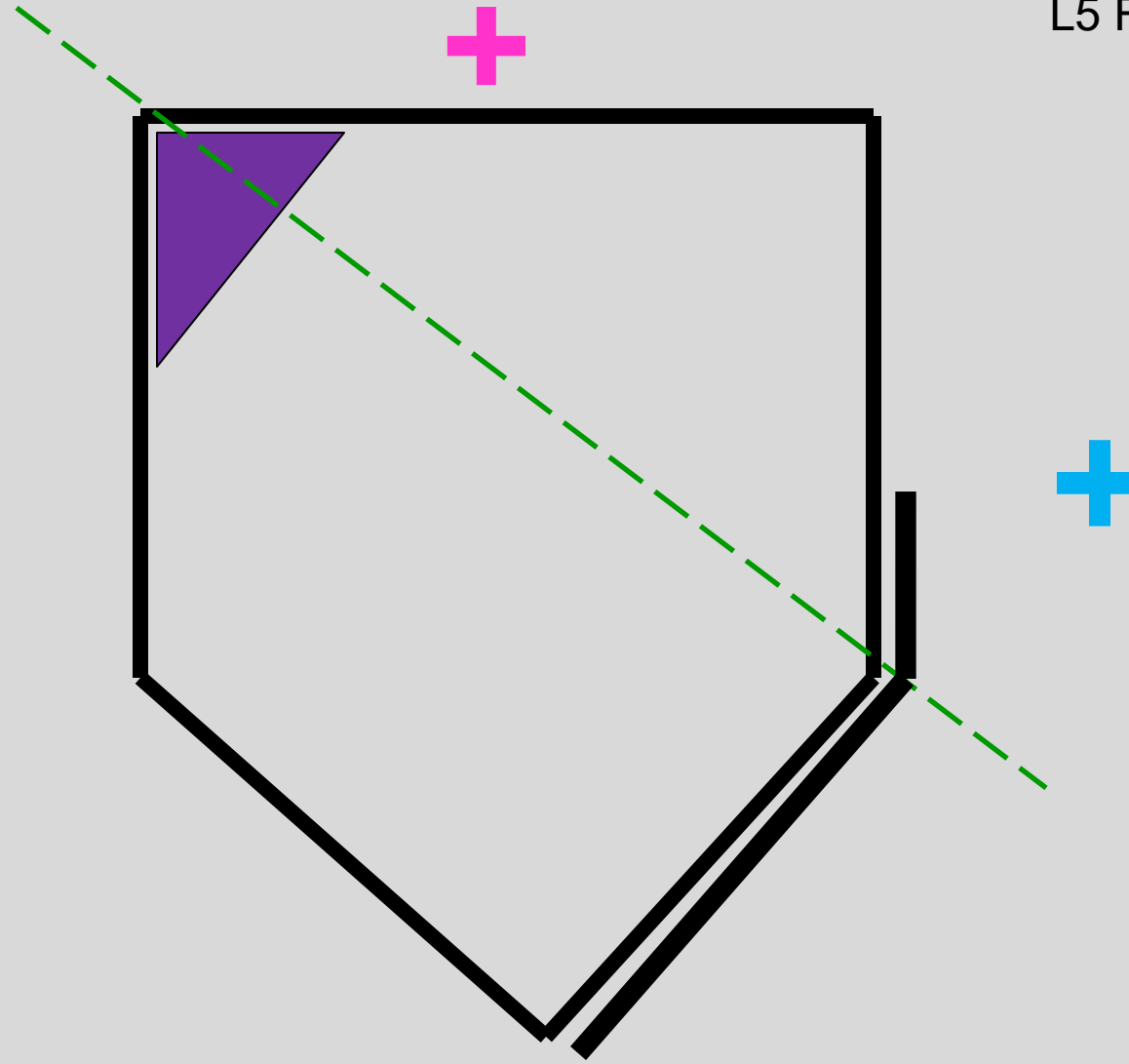


- L5 FRS left
- Seated Flexion???
- Spring test???
- Deep Sulcus???
- Post/inf ILA???

Work Slide # 11:

Right on Left Backward sacral torsion

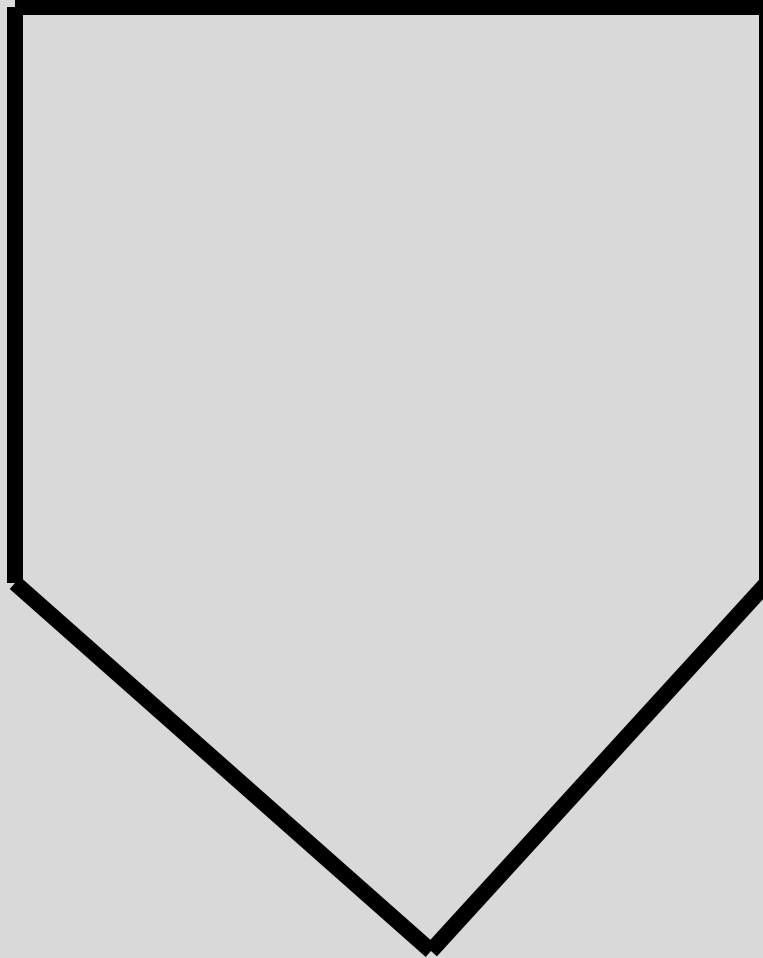
L5 FRS left



- L5 FRS left
- Seated Flexion + on R
- Spring test +
- Deep Sulcus on L
- Post/inf ILA on R

Work slide #12

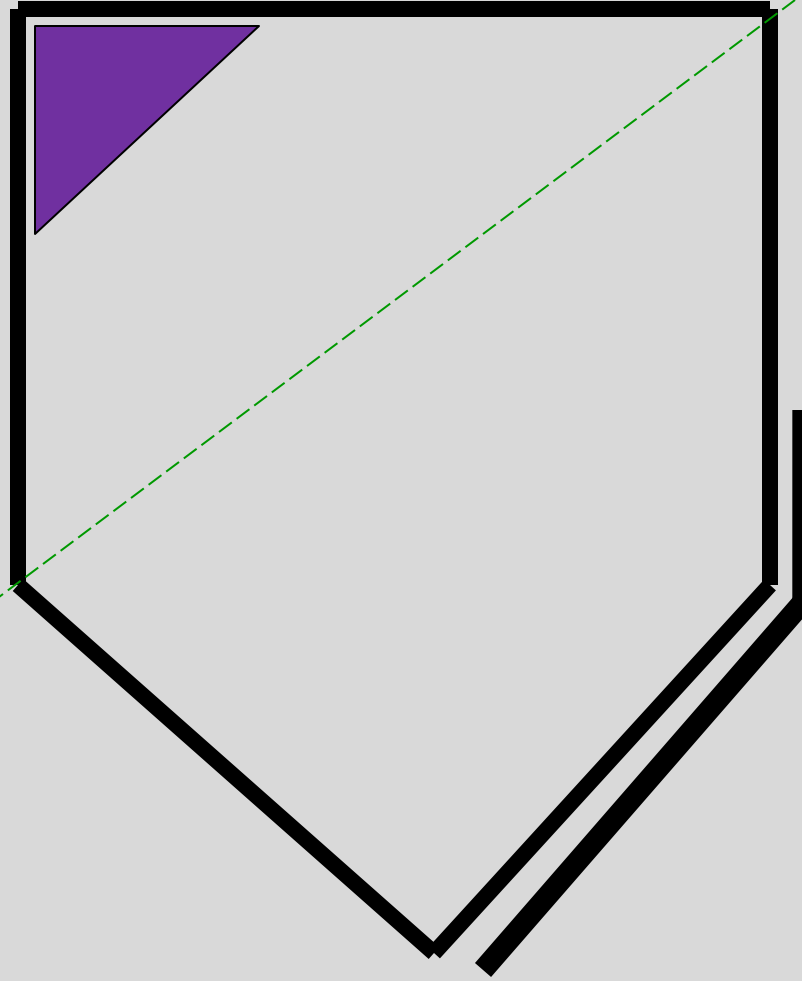
L5 N S right R left



- L3-5 N S right R left
- Spring test ????
- bbt / sphinx test ???
- Seated flexion test??
- Deep sulcus ???
- Post/Inf ILA ???



- L3-5 N S right R left
- Negative Spring test
- Neg. bbt / sphinx test
- L Seat. flexion test +
- Deep sulcus on left
- Posterior/Inferior ILA on the right



L5 N SR RL
(part of a
type I
dysfunction)

SELF ASSESSMENT CASES!!

- See if you can diagram out your sacral findings and make diagnoses for the following cases.
- If time permits, the cases will be discussed and worked out in lecture.

- Following each case question slide is a work slide WITHOUT the diagramming. Use this slide to work out your diagnosis. The following slide has the answer. The diagram answer is not there. Use the above slides for reference. If you answer incorrectly, retry diagramming and getting the correct answer.

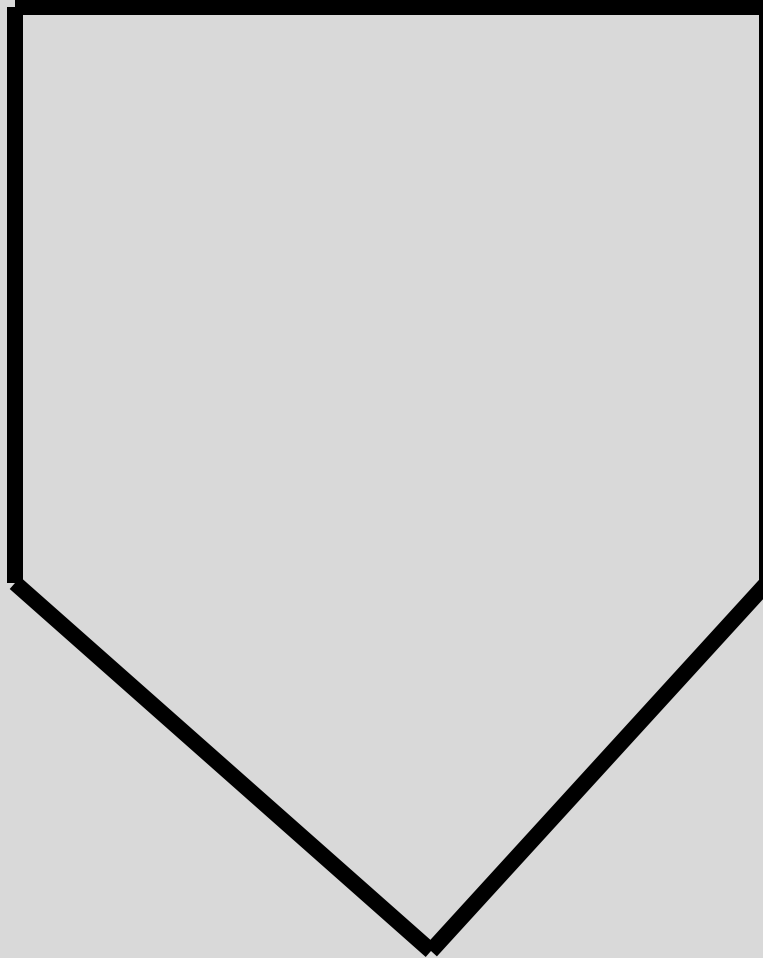
Self Assessment Question #1

A 24-year-old medical student falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the left, a negative spring test, a negative sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. L5 is rotated left. What is your sacral diagnosis?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

Self Assessment Question #1

**+ seated flex.
On L
Neg. sphinx test
Neg. Spring test
Deep sacral
sulcus on L
Post. ILA on R
L5 rotated Left**



Self Assessment Question #1

A 24-year-old medical student falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the left, a negative spring test, a negative sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. L5 is rotated left. What is your sacral diagnosis?

1. **R on R Forward Sacral Torsion**
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

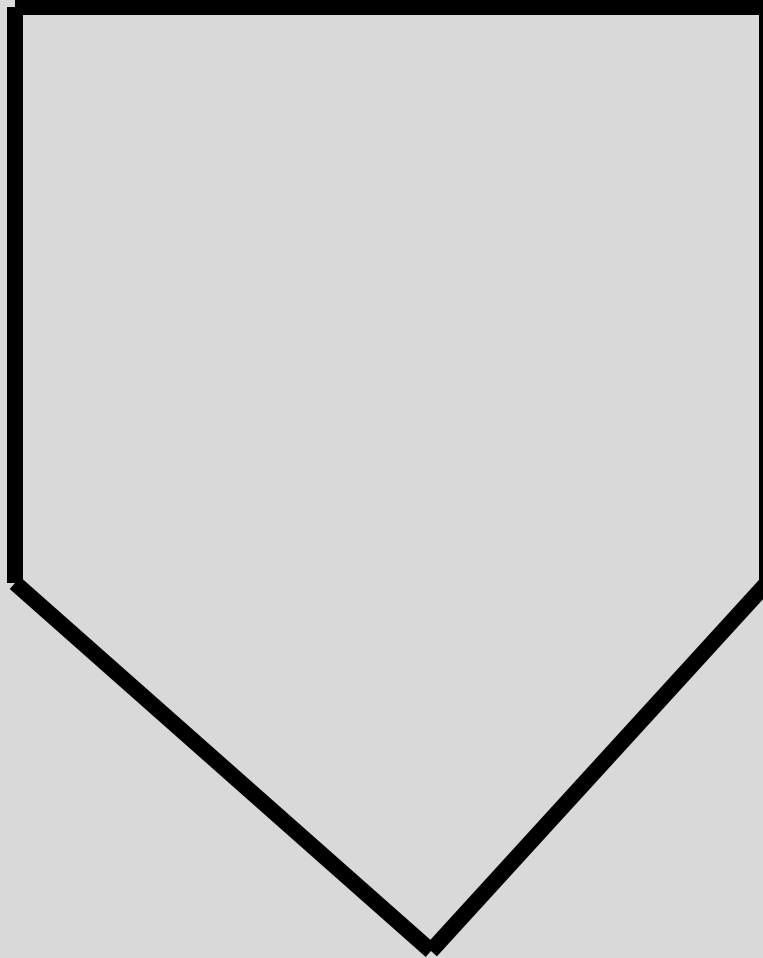
Self Assessment Question #2

A 35-year-old butcher falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the right, a positive spring test, a positive sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. L5 FRS left. What is your sacral diagnosis?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

Self Assessment Question #2

**+ seated flex. On
R
+ spring test
+ sphinx test
Deep sacral
sulcus on L
Post. ILA on R
L5 FRS left**



Self Assessment Question #2

A 35-year-old butcher falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the right, a positive spring test, a positive sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. L5 FRS left. What is your sacral diagnosis?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. **R on L Backward Sacral Torsion**
5. Right Unilateral Sacral Flexion

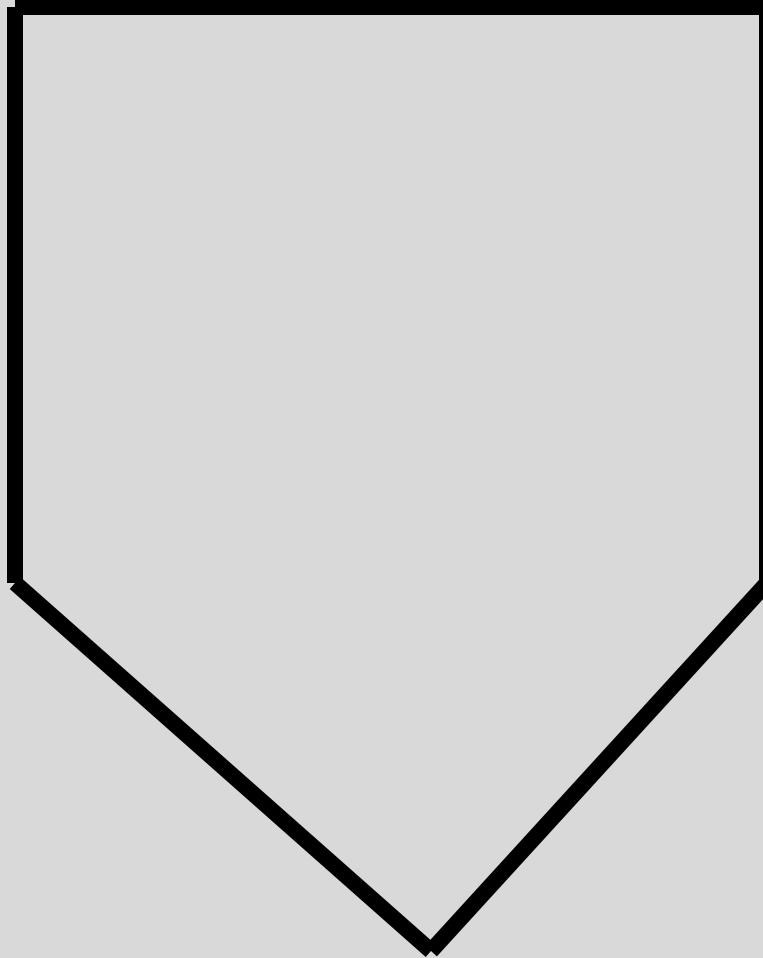
Self Assessment Question #3

A 24-year-old man had a heavy box fall out of his closet and twisted and caught it awkwardly. He presents to your office c/o low-back pain. On exam you find a positive seated flexion test on the right, a positive spring test, a shallow sacral sulcus on the right, an anterior-superior ILA on the right and L5 rotated right. What is your sacral diagnosis?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. Left unilateral Sacral Flexion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Extension

Self Assessment Question #3

**+ seated flex. On
R
Pos. Spring test
Pos. sphinx test
Shallow sacral
sulcus on R
Ant/Sup ILA on R**



Self Assessment Question #3

A 24-year-old man had a heavy box fall out of his closet and twisted and caught it awkwardly. He presents to your office c/o low-back pain. On exam you find a positive seated flexion test on the right, a positive spring test, a shallow sacral sulcus on the right, an anterior-superior ILA on the right and L5 rotated right. What is your sacral diagnosis?

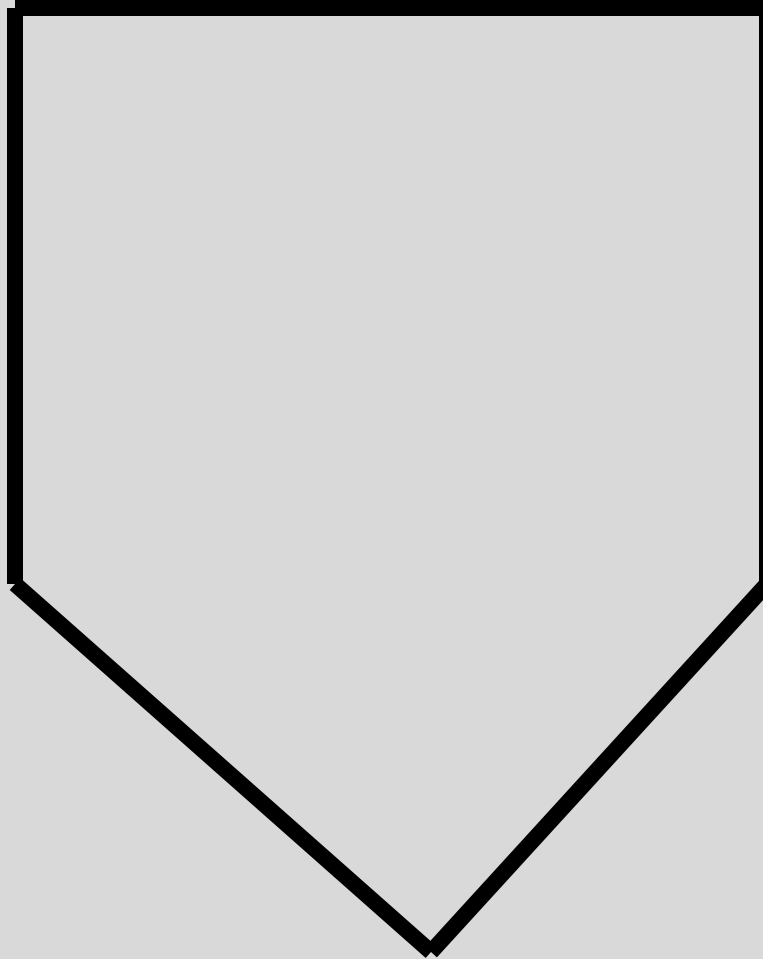
1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. Left unilateral Sacral Flexion
4. R on L Backward Sacral Torsion
5. **Right Unilateral Sacral Extension**

Self Assessment Question #4

A 24-year-old medical student falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the left, a positive standing flexion test on the left, a negative spring test, a negative sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. ASIS elevated on left, PSIS lower on the left, pubic tubercles are about even. L5 is rotated left. What is your sacral diagnosis?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

Self Assessment Question #4



**+ seated flex. On
L
Neg. sphinx test
Neg. Spring test
Deep sacral
sulcus on L
Post. ILA on R
L5 rotated left**

**+ standing flex on L
ASIS higher on L
PSIS lower on L
Pubic Tubercles =**

Self Assessment Question #4

A 24-year-old medical student falls on his buttocks and comes into your office c/o low-back pain. On exam you find a positive seated flexion test on the left, a positive standing flexion test on the left, a negative spring test, a negative sphinx test, a deep sacral sulcus on the left and a posterior ILA on the right. ASIS elevated on left, PSIS lower on the left, pubic tubercles are about even. L5 is rotated left. What is your sacral diagnosis?

- 1. R on R Forward Sacral Torsion (also a left posteriorly rotated innominate)**
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

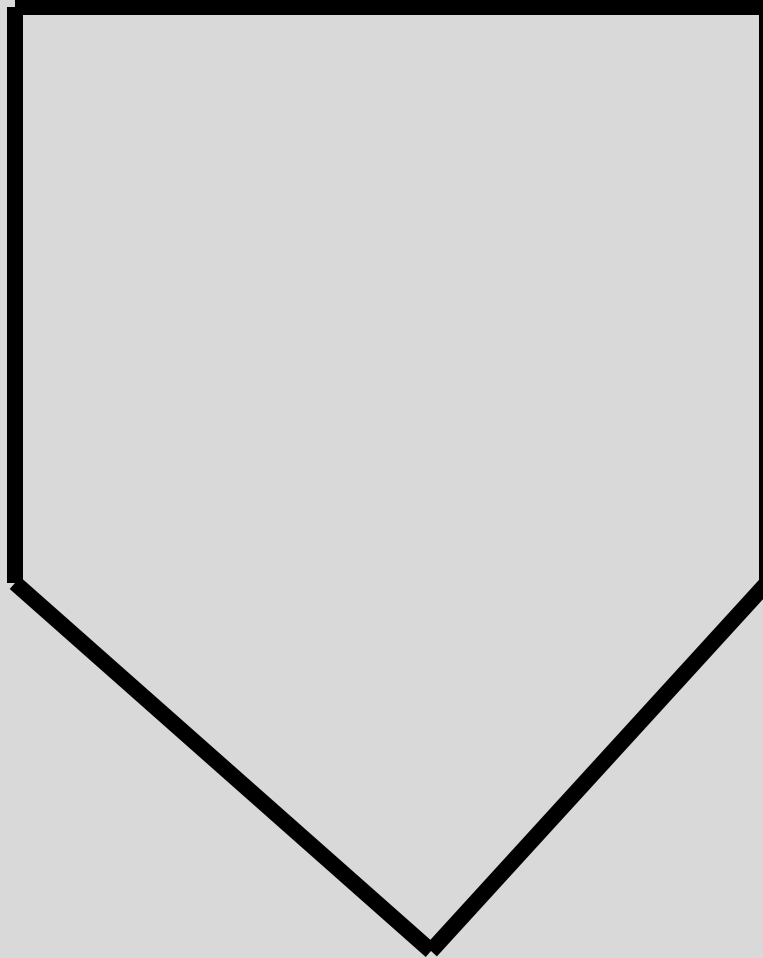
Self Assessment Question #5

A 44-year-old plumber falls on to his buttocks and comes into your office c/o low-back pain. L1-5 N sidebent left, rotated right. What is his sacral diagnosis most likely to be, based on the lumbar findings?

1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. L on L Forward Sacral Torsion
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

Self Assessment Question #5

seated flex. ???
sphinx test ???
spring test ???
Deep sacral sulcus
???
Post. ILA ???
L1-5 N S left R right



Self Assessment Question #5

A 44-year-old plumber falls on to his buttocks and comes into your office c/o low-back pain. L1-5 N sidebent left, rotated right. What is his sacral diagnosis most likely to be, based on the lumbar findings?

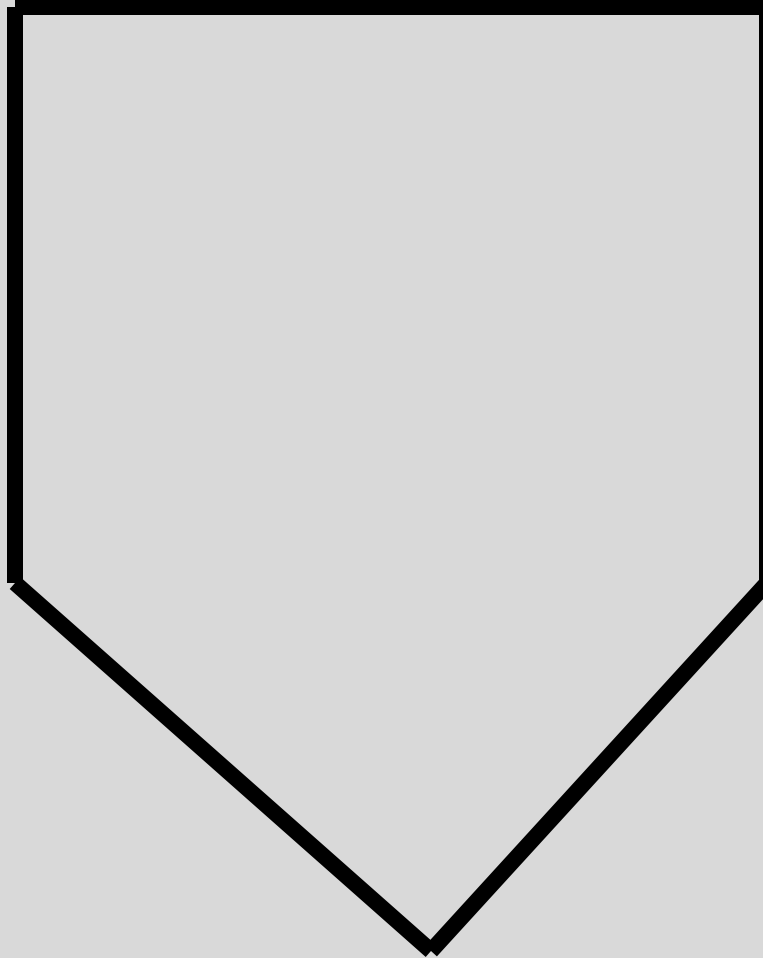
1. R on R Forward Sacral Torsion
2. L on R backward sacral torsion
3. **L on L Forward Sacral Torsion**
4. R on L Backward Sacral Torsion
5. Right Unilateral Sacral Flexion

Self Assessment Question #6

A 35-year-old song writer presents your office c/o low-back pain. On exam you find L5 extended, rotated and side-bent right. From this finding you make a likely sacral diagnosis. Which of the following would you expect to go along with your sacral diagnosis?

1. Negative Spring test
2. Posterior-inferior ILA on right
3. Left positive seated flexion test
4. Left Deep Sulcus
5. Left oblique axis

Self Assessment Question #6



A 35-year-old song writer presents to your office c/o low-back pain. On exam you find L5 extended, rotated and side-bent right. From this finding you make a likely sacral diagnosis. Which of the following would you expect to go along with your sacral diagnosis?

1. Negative Spring test
2. Posterior-inferior ILA on right
3. **Left positive seated flexion test**

(The dx is Left on Right Backward Sacral Torsion)

4. Left Deep Sulcus
5. Left oblique axis

Physical Findings and their Sacral Diagnoses

STATIC FINDINGS				TESTS			DIAGNOSIS	AXIS
ILA	Deep	Shallow	L5	Seated	Lumbar	Sphinx		
Posterior	Sulcus	Sulcus	Rotation	Flexion	Spring	Test		
Right	Right	Left		Right	Negative	Negative	Right Unilateral Shear	
Left	Left	Right		Left	Negative	Negative	Left Unilateral Shear	
Right	Left	Right	Right	Left	Negative	Negative	Right on Right (R on R) Rotation	Right Oblique
Right	Left	Right	Left	Left	Negative	Negative	Right on Right (R on R) Torsion	Right Oblique
Left	Right	Left	Left	Right	Negative	Negative	Left on Left (L on L) Rotation	Left Oblique
Left	Right	Left	Right	Right	Negative	Negative	Left on Left (L on L) Torsion	Left Oblique
Right	Left	Right	Right	Right	Positive	Positive	Right on Left (R on L) Rotation	Left Oblique
Right	Left	Right	Left	Right	Positive	Positive	Right on Left (R on L) Torsion	Left Oblique
Left	Right	Left	Left	Left	Positive	Positive	Left on Right (L on R) Rotation	Right Oblique
Left	Right	Left	Right	Left	Positive	Positive	Left on Right (L on R) Torsion	Right Oblique
Even	Bilateral			Equivocal	Negative	Negative	Bilateral Sacral Flexion	Transverse
Even		Bilateral		Equivocal	Equivocal		Bilateral Sacral Extension	Transverse

DiGiovanna, EL & Schiowitz, S, An Osteopathic Approach to Dx & Tx, Lippincott W &W
Table 60-1

Algorithm of Sacral Diagnosis:

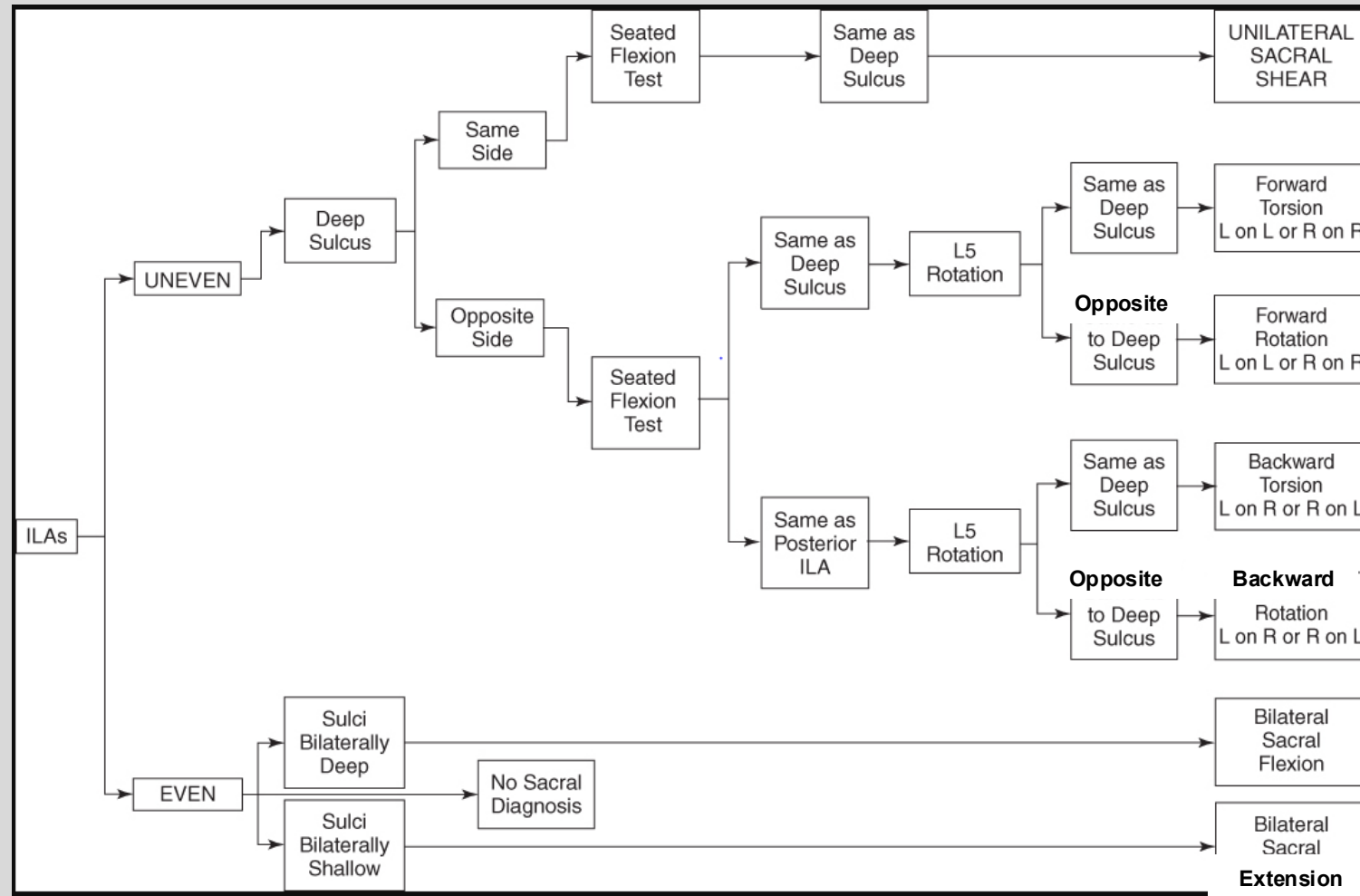


Table modified from: DiGiovanna 4e – Table 60-13

Summary Slide

- Static physical exam findings of the sacral region will be used to make the diagnosis of sacral somatic dysfunctions
- Diagraming your physical findings will help you in making the somatic dysfunction diagnosis

References

Core Reference:

- Dowling D. “Evaluation of the Sacrum” in:
DiGiovanna, EL, Amen, CJ, and Burns, DK. An Osteopathic Approach to Diagnosis & Treatment 4e, Lippincott W & W, 2020, Chapter 60
<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=248814152&bookid=2969>

Other References:

- Heinking, KP, Giusti, ER, Hensel, KL. *Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research 5th Edition*: Ch 38
<https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=261502266&bookid=3456#261502267>
- Mitchell, Fred L., Jr. The Muscle Energy Manual, vol 3 (1999), p. 21-51 & 121-150.
 - Greenman, P. Principles of Manual Medicine, 2017 5th ed Figures 17.3 to 17.51

Lecture Feedback Form:

[https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWY
XREL4R](https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R)